

MODERN DIGITAL TECHNOLOGIES AND AUTOMATIC CONTROL SYSTEMS IN LOGISTICS COMPANIES - DIGITAL LOGISTICS

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RESUME

The article discusses the relevance and importance of introducing modern digital technologies into the activities of logistics companies. Due to the implemented innovations, digitalization is one of the most important tools for survival in the new conditions. Today, business is moving from global digitalization to global adaptation. Businesses will have to solve technical issues (technique, components, software); to develop new and change existing IT services, strengthen physical, economic, organizational, personnel, information, legal security. Enterprises have the opportunity to increase their competitiveness and be customer-oriented in the logistics market. The article also provides examples of various automated control systems in individual logistics processes and their use in practice. The use of modern information technologies and intelligent transportation management systems is a key area for improving the process of cargo delivery.

Digitalization greatly simplifies the transportation of goods, provides flexibility in working with customers and allows one to pay in any currency using various payment systems. In addition to electronic document management and simplification of the communication process, digital logistics in the future involves the use of unmanned vehicles, drones. Such facilities provide faster delivery of goods or communication in remote areas when the roads are congested.

Key words: digital technologies, digitalization, increasing competitiveness and focus on customer, digital logistics, logistics market, intelligent transportation management systems, information technology, multimodal transportation systems.

INTRODUCTION

The logistics market is constantly evolving. In order to remain competitive, it is necessary to constantly be oriented on customer, innovation and progress. Businesses use a variety of methods. This is the introduction of the latest marketing methods, innovative developments in the organization of production and logistics activities to popularize and promote goods and services.

In the modern world, digitalization of processes is taking place in almost all spheres of human activity. The logistics market is not an exception as well. In the market of logistics services, systems of interaction with customers are constantly changing, and, of course, new ones are being formed and created. One of the aspects of the company's development is the expansion of competitiveness. In the field of logistics, this is possible only with the introduction of innovations and an intelligent marketing approach.

BASIC TEXT

Logistics represents a fairly large segment, an important and integral part of the global economy and the market of any country. Logistics planners accompany any market development process. Most of the large companies operating in the logistics industry invest most of their material resources in research and the latest technology, which in turn gives them a competitive advantage in the logistics market. Based on the basic concept of transport logistics - to deliver the goods to the place and time of need, with optimal delivery costs, enterprises are currently using all possible products of technical progress that reduce logistics costs or improve other performance indicators of the company.

Digital logistics, which is based on the use of modern information

technologies and intelligent transportation management systems, is a key area for improving the process of delivering goods from the sender to the recipient in compliance with all the principles of logistics, which allows automating the tasks of transport companies. For example, a system the purpose of which is to manage terminals in ports automates TMS (Terminal Management System) cargo in the port, reduces the time for operations and errors caused by human factors. In addition, the presented TMS system is used to automate the process of storing various cargoes in ports and terminals. Another system designed for warehouse management is WMS (Warehouse Management System), which is used in enterprises that specialize mainly in warehouse activities. The mentioned system automates the management of warehouse business processes of the enterprise. The functionality of this system allows one to centrally perform warehouse operations using software and also guarantees data transparency throughout the supply chain.

In practice, companies use the information system ARM - ARM (workstation) - information and technical system that collects, stores and processes data, as well as automated management of various business processes of employees based on specialization. Information about a new shipment request, which will be received by the customer's department from the customer and entered into the ACA system, will be instantly available to all departments of the company.

In practice, companies use the information system - Automated workplace AWP (automated workplace)- information and technical system that collects, stores and processes data, as well as automated management of various business processes of employees based on specialization. Information about a new shipment request, which will be received by the customer's department from the client and entered into the AWP system, will be instantly available to all departments of the company. Upon receiving a new order, an employee of the logistics department immediately begins to work on it and arranges a sea freight. All conversations between logistics and customer department employees will also go through the same system.

When the time comes for customs clearance of cargo in the port, the customs service sees the specified order and starts working on it without additional notice to the client's managers.

Financial sub-division also has an access to the information system automated workplace (AWP) which allows one to financially close all completed deliveries in a timely manner and without unnecessary exchange of mail messages.

The mentioned system allows department heads to control the actions of their employees. It also allows important transportation statistics data to be collected for further analysis. Automated workplace -AWP significantly reduces the level of employees' errors.

New solutions in logistics using digital technologies are often presented at various international exhibitions in the field of transport and logistics services and technologies, transport and logistics services, warehouse equipment and technologies.

In addition, various news portals reflect all the latest technologies and changes that have taken place in recent years. A number of innovations in warehouse logistics are especially highlighted.

Nowadays EDI (Electronic Data Interchange) technologies are distinguished as one of the most developing and promising areas. It represents the exchange of various information data through digital technologies. Thanks to the mentioned EDI technologies, exchanging electronic messages between business partners in a unified form became possible. The objective of the presented technology is the exchange of electronic data in a common format between computer systems. Since 2018, EDI has been implemented in almost all warehouse systems.

Presented technologies are mostly used in the exchange of transport and warehouse documents such as Delivery Order, inventory lists, warehouse receipts, shipment notices, etc. After the first steps of actively introducing EDI technology into the warehouse process management system, it revealed the following benefits. Namely:

- Smoothness and transparency of information flow between different computer systems;

- Compatibility of documentation;
- General transparency of the entire process of work between the

various elements of the logistics system in the field of data exchange;

It can be assumed that this technology encourages the development of platforms that integrate the entire workflow associated with the transportation of goods, regardless of the type of transportation of the goods. The creation of such a platform will facilitate the registration of cargo transportation, in particular when organizing multimodal cargo transportation. Given the complexity and specifics of multimodal transportation, such platform would give us the opportunity to bring this process to a new level.

The next direction of development is related to the introduction of the concept of seamlessly integrated channels. This concept is characterized by connecting the buyer with the seller. It involves the integration and coordination of all channels of communication with the client, which is well reflected in the client-centered approach. "Omni-channel" or in other words "end-to-end analytics", "penetrative analytics" help the service or product provider to bring the client to the final goal - to buy or choose a service. According to Analysts, by 2021, 85% of businesses has moved online. Warehouse operators and logistics companies must keep up with development trends, as this will lead to a loss of competitiveness. According to statistics, most warehouses plan to significantly increase investment in the development of information technology. This type of distribution is increasing in parallel with EDI technologies.

CONCLUSION

Having considered some of the main modern digital technologies in the logistics market, we can conclude that the pace of digitalization in this area is very high. This is due to the high competition in the market.

But the level of development of digital technologies for organizing multimodal transportation is not high enough. This is due to many aspects of this mentioned process. For example, in order to fa-

cilitate multimodal transportation, a unified approach to document flow and its execution is required, but this is not yet possible, since not all issues of organizing delivery on multimodal principles are regulated by law.

In order to increase orientation and competitiveness on the customer, enterprises need to make greater use of the capabilities of new software products in logistics based on digital technologies, which will allow logistics business processes to be optimized and to increase their share of services on the market.

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Digitization is one of the most important tools for survival in the new world. Today, business is moving from global digitalization to global adaptation. Business will have to solve the technical issues (hardware, components, software); develop new and replace existing IT services, enhance physical, economic, organizational, personnel, information, legal security. The article considers the relevance and importance of introducing modern digital technologies into the activities of logistics companies. Thanks to innovations, many companies have the opportunity to increase their competitiveness and client-oriented approach in the logistics market. The article also contains examples of various automated control systems in certain logistics processes and their application in practice. The use of modern information technologies and intelligent transport management systems is a key area for improving the delivery of goods.

The use of modern information technologies and intelligent transport management systems is a key area for improving the delivery of goods so the article considers the relevance and importance of introducing modern digital technologies into the activities

of logistics companies. The article also contains examples of various automated control systems in certain logistics processes and their application in practice. Digitalization makes the transportation of goods much simpler, provides flexibility in working with customers and allows one to pay in any currency through various payment systems. In addition to simplifying electronic document management and communication process, digital logistics in the future is considered to use unmanned vehicles and drones. Such facilities provide faster delivery of goods or communication to remote areas when roads are congested.

Key words: logistics, digitalization, information technology, automated logistics, digital logistics and technologies, increasing competitiveness and client-oriented approach, logistics market, intelligent transport management systems, multimodal cargo transportation.

თანამედროვე ციფრული ტექნოლოგიები და ავტომატური მართვის სისტემები ლოგისტიკურ კომპანიებში - ციფრული ლოგისტიკა

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რეზიუმე

განხილულია თანამედროვე ციფრული ტექნოლოგიების დანერგვის აქტუალობა და მნიშვნელობა ლოგისტიკური კომპანიების საქმიანობაში. დანერგილი ინოვაციების წყალობით, დიგიტალიზაცია ახალ პირობებში გადარჩენის ერთ-ერთი ყველაზე მნიშვნელოვანი ინსტრუმენტია. დღეს ბიზნესი გლობალური დიგიტალიზაციიდან გლობალურ ადაპტაციაზე გადადის. ბიზნესს მოუწევს ტექნიკური საკითხების გადაჭრა (ტექნიკა, კომპონენტები, პროგრამული უზრუნველყოფა); განავითაროს ახალი და

შეცვალოს არსებული IT სერვისები, გააძლიეროს ფიზიკური, ეკონომიკური, ორგანიზაციული, საკადრო, საინფორმაციო, იურიდიული უსაფრთხოება. საწარმოებს აქვთ შესაძლებლობა გაზარდონ კონკურენტუნარიანობა და იყვნენ მომხმარებელზე ორიენტირებული ლოგისტიკის ბაზარზე. სტატიაში ასევე მოცემულია სხვადასხვა ავტომატური მართვის სისტემების მაგალითები გარკვეულ ლოგისტიკურ პროცესებში და მათი გამოყენება პრაქტიკაში. თანამედროვე საინფორმაციო ტექნოლოგიებისა და ტრანსპორტის მართვის ინტელექტუალური სისტემების გამოყენება საკვანძო მიმართულებაა ტვირთების მიწოდების პროცესის გასაუმჯობესებლად.

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